

EXPERIMENT 2

Question 1:

Objective: Write a MATLAB program to calculate:

- a) Summation of a number k from 0 to 100.
- b) Summation of a number k^2 from 0 to 100.
- c) Summation of a number $1/k$ from 1 to 100.
- d) Summation of a number $1/k^2$ from 1 to 100.

Software Required:

MATLAB software.

Theory:

Element wise Operation: If we apply “.” before any operation then that operation is operated on every element of array separately.

sum(A): If A is a vector, returns the sum of the elements of the vector.

Procedure:

Open MATLAB in your PC.

Open a new script.

Write the code as mentioned in the next heading in the script.

Save the script.

Run the script.

Desired result will be displayed on the command window.

Code:

```
N= 0:100;
a = sum(N);
fprintf('Sum of n numbers =')
display(a)
b = sum(N.^2);
fprintf('Sum of square of n numbers =')
display(b)
N= 1:100;
c = sum(1./N);
```

```
fprintf('Sum of reciprocal of n numbers =')  
display(c)  
d= sum(1./N.^2);  
fprintf('Sum of square of reciprocal of n numbers =')  
display(d)
```

Expeted Outcome:

Command Window

New to MATLAB? See resources for [Getting Started](#).

```
Sum of n numbers=  
a =  
  
    5050  
  
Sum of square of n numbers=  
b =  
  
   338350  
  
Sum of reciprocal numbers=  
c =  
  
    5.1874  
  
Sum of square of reciprocal numbers=  
d =  
  
    1.6350  
  
>>
```

Conclusion:

In this experiment, the motive was to introduce the element wise operations on array using MATLAB.

Question 2:

Objective: Write a MATLAB program to break a function into odd and even parts.

Software Required:

MATLAB software.

Theory:

Even and odd signals bear some important symmetry properties. Under reversal of independent variable, these signals either remain the same (even signal) or get reflected or flipped (odd signal) about the horizontal axis. Equations or definitions (1) and (2) mathematically express these properties for both continuous and discrete time cases.

$$\text{Even Signals: } x(t) = x(-t), x[n] = x[-n] \quad (1)$$

$$\text{Odd Signals: } x(t) = -x(-t), x[n] = -x[-n] \quad (2)$$

A signal $x(t)$ can be decomposed into even and odd parts as expressed by the equations (3) and (4)

$$x_e(t) = \frac{1}{2}\{x(t) + x(-t)\} \quad (3)$$

$$x_o(t) = \frac{1}{2}\{x(t) - x(-t)\} \quad (4)$$

Where $x(t) = x_e(t) + x_o(t)$

Procedure:

Open MATLAB in your PC.

Open a new script.

Write the code as mentioned in the next heading in the script.

Save the script.

Run the script.

Desired result will be displayed on the command window.

Code:

```
clc clear all
close all
x=0:1:10
a= exp(x)
b= exp(-x)
odd=(a-b)/2
even = (a+b)/2
```

Expected Outcome:

```
Command Window
New to MATLAB? See resources for Getting Started.

a =
    1.0e+04 *
    0.0001    0.0003    0.0007    0.0020    0.0055    0.0148    0.0403    0.1097    0.2981    0.8103    2.2026

b =
    1.0000    0.3679    0.1353    0.0498    0.0183    0.0067    0.0025    0.0009    0.0003    0.0001    0.0000

odd =
    1.0e+04 *
         0    0.0001    0.0004    0.0010    0.0027    0.0074    0.0202    0.0548    0.1490    0.4052    1.1013

even =
    1.0e+04 *
    0.0001    0.0002    0.0004    0.0010    0.0027    0.0074    0.0202    0.0548    0.1490    0.4052    1.1013

>>
```

Conclusion:

In this experiment, the motive was to understand MATLAB tool is used to break a signal or function in even and odd parts.